

Cost-Effective, High Performance Heterogeneous Integration for 6.4T and Beyond 224Gbps/lane Co-Packaged-Optical Engines for AI/ML & Data Center

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ABSTRACT

Heterogeneous integration (HI) is a highly promising approach to integrate photonic and electronic chips to form optical engines (OE) which address the crucial I/O bottleneck in hyperscale Data Centre (DC) switches and AI/ML clusters in high performance computing (HPC). Cost, performance, and manufacturability are key challenges for HI. In this paper we present the design, process-integration, test-results for a novel Fan-out Wafer Level Package (FOWLP) approach to realize cost-effective, volume-manufacturable, high-performance OEs for 1.6T/6.4T and beyond.

Keywords: Heterogeneous Integration, Optical Engine, Fan-Out Wafer Level Package, Co-Packaged Optics

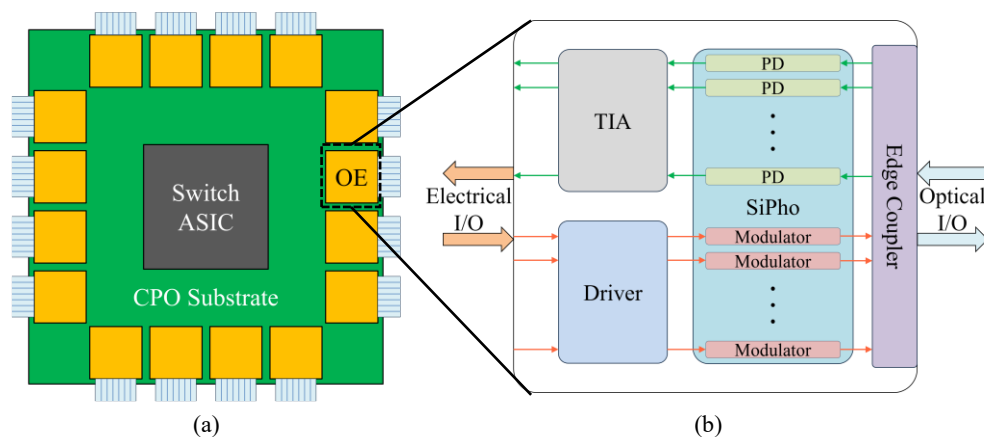


Figure 1. Co-packaged optics assembly with sixteen OEs around the perimeter. (b) Functional block diagram for optical transceiver module (optical engine).

1. INTRODUCTION

Exponential growth in AI/ML model complexity is driving the rapidly increasing need for optical connectivity in AI/ML clusters for HPC, and hyperscale DC. The top-of-rack switch bandwidth in hyperscale DCs is expected to reach >51.2Tbps by 2024 [1], where large form-factor conventional pluggable optical transceivers with 10s of cm long switch-to-transceiver Cu interconnects will not be able to meet the power and bandwidth requirements of optical to electrical conversion and vice-versa. Co-Packaged Optics (CPO), as shown in Figure 1(a), with small form-factor OEs and switch-ASIC packaged on the same substrate/interposer enables extremely short interconnects (10s of mm) and thus, is a vital solution for future DCs. Such OEs convert optical signals to electrical signals and vice versa with Photonic IC (PIC) for the receive and

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transmit of optical signals as shown in Figure 1(b). Electronic ICs (EICs), TIA (Trans-Impedance Amplifier) and Driver amplify the weak signals received from the fiber, and drive PIC modulators respectively. There is a crucial need for manufacturable HI packaging which can perform the challenging task of integrating PIC and EICs into a compact OE. In this paper we demonstrate a cost-effective HI approach using FOWLP for high performance OE.

2. HETEROGENEOUS INTEGRATION PLATFORMS FOR OPTICAL ENGINES

Three commonly used HI approaches for OE are: 2.5D interposer [2], 3D-TSV [1], and FOWLP [4] [5] as shown in Figure 2. In 2.5D approach, EICs and PIC are flip-chip mounted on top of an interposer or substrate. The interconnects between EIC and PIC are through μ bumps and interposer Re-Distribution Layers (RDL). With 3D-TSV approach, the TSV is formed with keep-out zones in the PIC with the EICs flip-chipped on top of the PIC using μ bumps. In FOWLP approach, the PIC is embedded in a mold compound with Through-Mold-Via (TMV) formed in mold. EICs are similarly flip-chipped on top. OE-switch I/O interconnects for three approaches are routed through TIVs (Through-Interposer-Vias), TSVs, and TMVs respectively to package substrate.

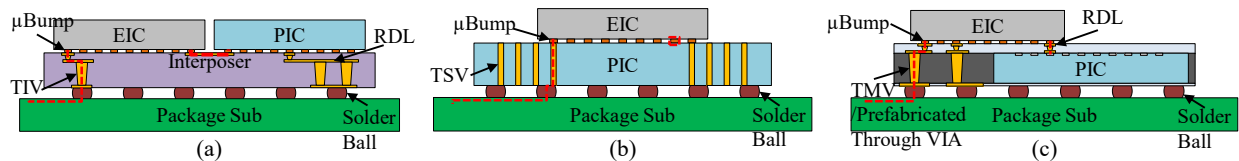


Figure 2. HI platforms for OE packaging. Not drawn to scale. Red dashed lines indicate the signal paths. (a) 2.5D integration of both EIC and PIC flip-chip connected on an interposer. (b) 3D integration of EIC flip-chip connected on top of PIC with TSV. (c) FOWLP integration of PIC embedded in mold and EIC flip-chip connected on top.

Table 1 compares the three approaches. 3D-TSV and FOWLP can have the shortest EIC-PIC interconnects with the direct μ bump connections, while 2.5D interposer requires an additional section of interposer RDL. All three approaches require through-vias for OE-Switch interconnects. For 3D-TSV approach, high loss and noise coupling due to the high capacitance between TSV and silicon substrate can occur at high frequency because of silicon’s conductive properties [6]. Due to the 3D stack-up of EIC on PIC, 3D-TSV and FOWLP have smaller package areas compared to 2.5D interposer. On one hand, 3D-TSV can have higher I/O density and in turn smaller package area by using fine-pitch TSVs and smaller diameters. On the other hand, the keep-out zones for TSV may increase the PIC size. Fine pitch prefabricated through vias with good performance like Through-Glass-Vias (TGV) can be embedded into mold of FOWLP to enable a higher I/O density. Because of the high I/O density, 3D-TSV and FOWLP can be scaled to higher bandwidth with more EICs and higher number of I/O. FOWLP has lower fabrication cost of 50% savings compared to 3D-TSV as shown in Figure 3 (a), and faster supply chain compared to 2.5D interposer and 3D-TSV approaches as compared in Figure 3 (b). Furthermore, only PIC Known Good Dies (KGDs) are selected for the FOWLP processing. In comparison, the entire PIC wafer undergoes

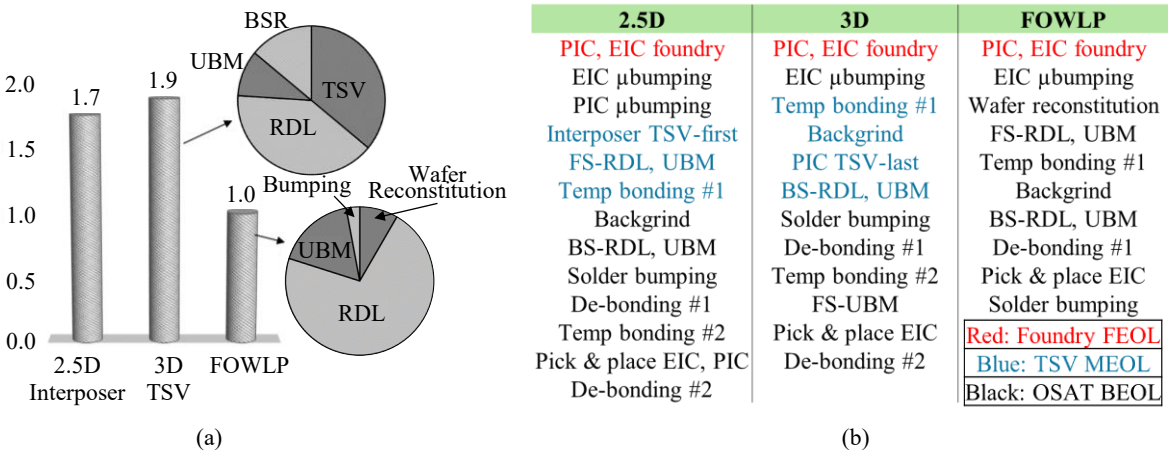


Figure 3. (a) The cost model comparison between 2.5D interposer, 3D-TSV, and FOWLP. (b) Key HI manufacturing steps for 2.5D interposer, 3D-TSV, and FOWLP OE packages.

Table 1. Comparison of three commonly used heterogenous integration (HI) platforms for optical engines.

	EIC-PIC Interconnect	Bandwidth Scalability	Area	Cost	Supply Chain
2.5D Interposer	--	-	--	--	--
3D TSV	+	+	++	--	-
FOWLP	+	++	+	++	+

the TSV process for 3D-TSV. To summarize, FOWLP is a cost-effective platform with low interconnect loss, small form-factor, high I/O density, and high scalability, making it an extremely promising solution for high bandwidth OEs.

3. OE PROCESS INTEGRATION AND ASSEMBLY

3.1 FOWLP Process Flow

A mold-first FOWLP process flow with the key steps shown in Figure 4 is employed for integrating the OE package. The process begins with reconstituting the molded wafer from silicon dies. The PIC is picked and placed face down on the adhesive mold plate, followed by compression molding to encapsulate it in resin. After reconstitution, the wafers are separated from the mold plate and proceed for frontside two RDL metal layers, Under-Bump-Metallization (UBM) layer, and dielectric processing. Following front-side processing, the wafer thickness is reduced to 200~300 μ m or less through back grinding before TMV and backside RDL processing. TMVs, linking frontside to backside, are made by means of laser drilling and copper plating. The backside RDL includes one metal layer and UBM. Figure 5 shows the pictures of the fabricated OE package. As an alternative for TMVs, a prefabricated substrate based through via can also be embedded in mold together with PIC as shown in Figure 5 (b) and (c). The process flow is designed to facilitate both vertical and edge optical coupling. Figure 5 (d) and (e) show the impact of the specially designed silicon buffer structure to prevent mold seepage into the deep trench and optical I/O region. With the buffer structure, a clean optical I/O region is obtained as seen in Figure 5 (d) comparing to Figure 5 (e) without the buffer structure.

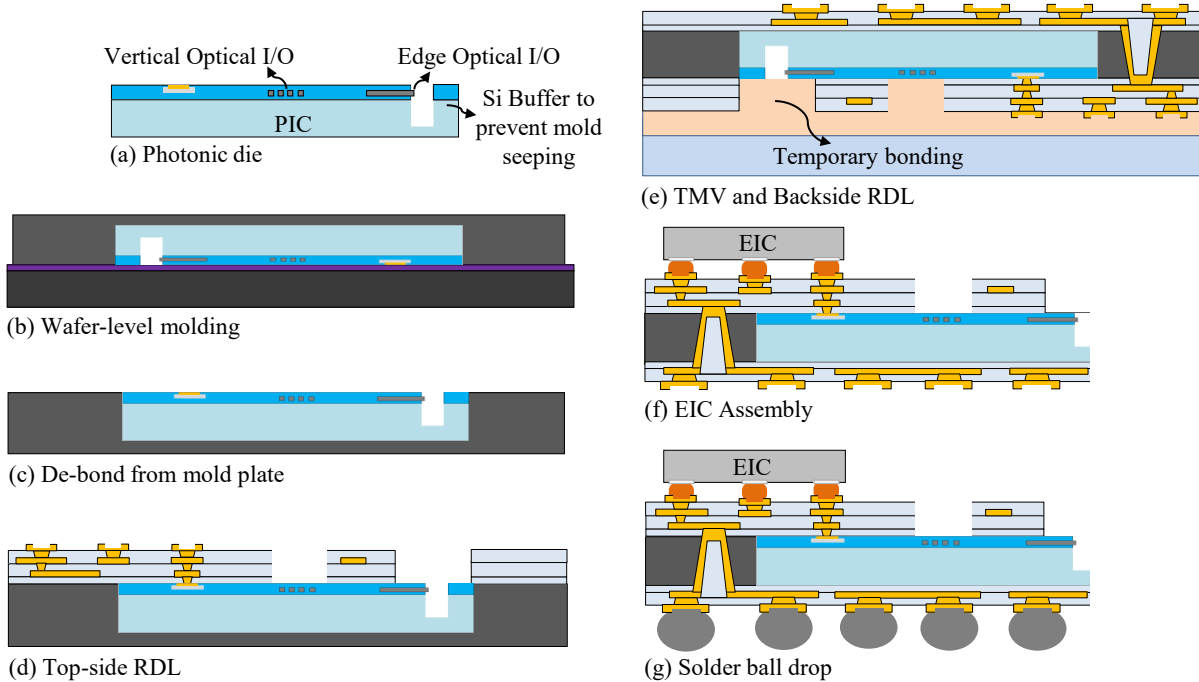


Figure 4. FOWLP process flow of the OE package

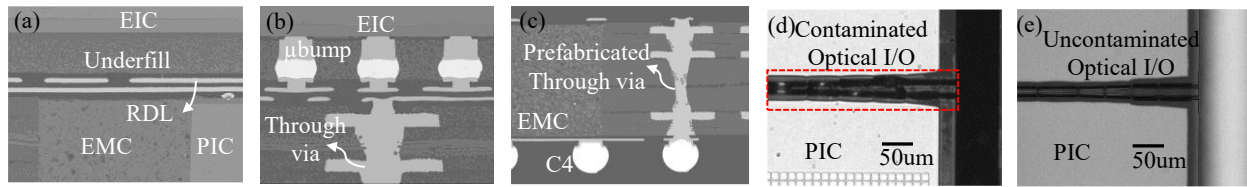


Figure 5. (a) Cross-sectional image showing the daisy chain structures on PIC. (b) Cross-sectional view of EIC flip-chip mounted on top and connected through μ bump to frontside RDL. (c) Cross-sectional view of the C4 bumps and embedded prefabricated substrate with through vias. (d) Image from a sample without the buffer structure, clearly showing mold compound seepage into the trench and Optical I/O area. (e) Image showing a clean Optical I/O area which is free from contamination due to the effective silicon buffer structure.

3.2 Wafer Monitor Data (Optical & Electrical)

Wafer level optical test was done through vertical optical I/O as shown in Figure 6 (b). Four PIC vertical optical I/O on nine OE packages were measured and constant optical insertion losses were obtained as shown in Figure 6 (c) except one which was affected by a particle on the coupler. After FOWLP packaging, an optical coupling loss of less than 2 dB/facet was obtained (in the absence of index-matching epoxy) as shown in Figure 6 (d). Daisy chain structures were used to study integration integrity by linking front-side RDL, micro vias, TMVs, and backside RDL. Wafer-level analysis confirms uninterrupted connectivity across all daisy chains without open defects, as depicted in Figure 7.

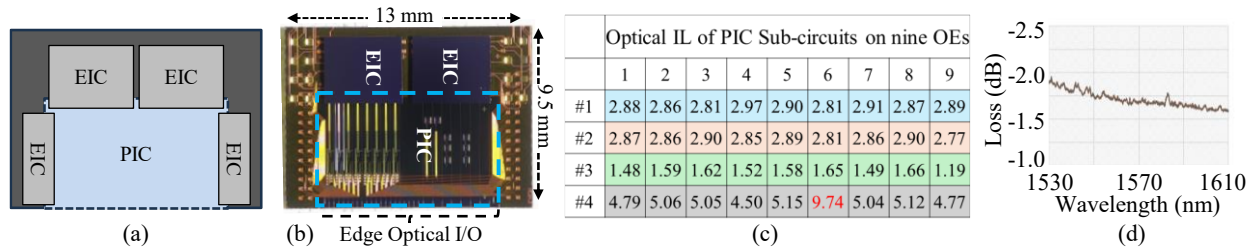


Figure 6. (a) 1.6 Tbps OE package schematic. (b) An OE package with 2 EICs assembled. (c) Wafer level optical test through Vertical Optical I/O. (d) Measurement of Fiber to Edge Optical I/O coupling loss after FOWLP packaging.

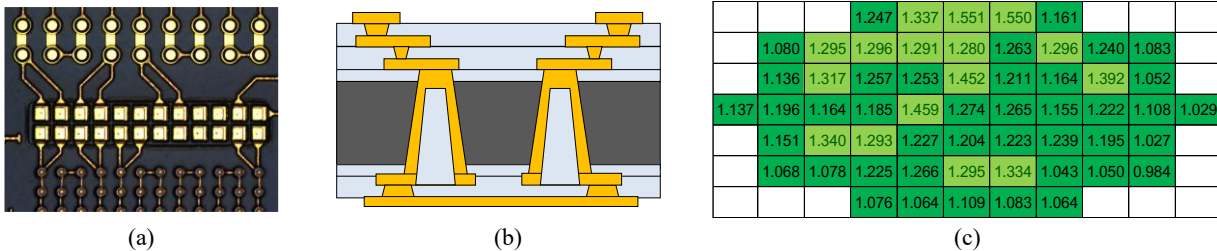


Figure 7. (a) Daisy chains that connect frontside UBM and RDLs to backside RDLs through TMV. (b) Daisy chain cross-sectional view. (c) Wafer-level DC test results of the daisy chain interconnects.

4. 1.6 TBPS OE PACKAGE DESIGN AND PERFORMANCE

4.1 OE Package Design and Performance

Figure 6 (a) shows the schematic of the 1.6 Tbps OE package with 4 EICs (2 TIAs and 2 drivers). The OE package has a dimension of 13 mm $\times$ 9.5 mm. 224 Gbps PAM4 signal was applied to drive a silicon photonic Traveling Wave Mach-Zehnder Modulator (TWMZM) through full routings in substrate and FOWLP. With 21-tap FFE equalization at the receive side, clear eye openings can be observed as shown in Figure 8 achieving excellent TDECQ of 2.25 dB.

4.2 Interconnects Performance

Figure 9 (a) shows the μ bump direct connection between EIC and PIC. The total signal path has a length of 126 μ m. The structure is simulated to have a differential insertion loss (IL) and return loss (RL) of 0.14 dB and 15.54 dB respectively

at 56 GHz as shown in Figure 9 (b). Figure 10 shows the fabricated TMV structure and the measured S-parameters. The structure was measured using double-side probing [7] from DC to 50 GHz. The structure is measured to have a differential IL of 0.18 dB and RL of 17 dB at 28 GHz. The double-side probing measurements were also applied to other type of through vias. Figure 11 shows the cross-sectional view of the fabricated TMV and prefabricated substrate through vias embedded in mold. Figure 11 (c) compares the IL (results normalized to 100 μm) of four types of through vias: TMV, substrate based, TGV, and TSV.

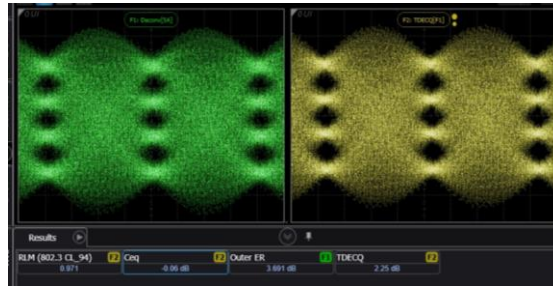


Figure 8. Optical transmit eye diagram measured at receive side for 224 Gbps PAM4 signal after 21 FFE receive-side taps.

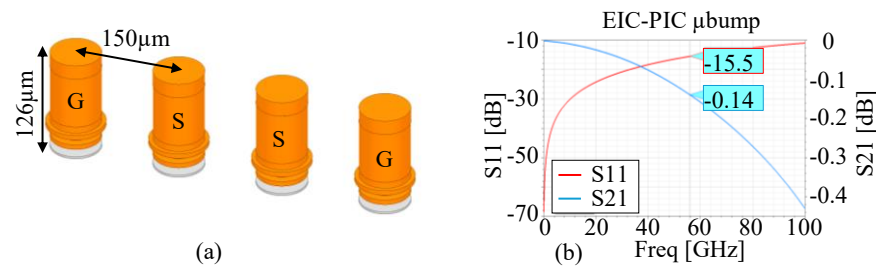


Figure 9. (a) μ bump interconnects between EIC and PIC. (b) Simulated differential insertion loss and return loss from DC to 100 GHz.

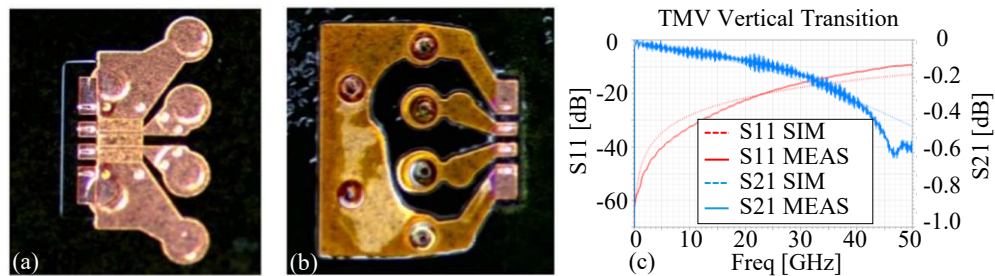


Figure 10. (a) Frontside of the fabricated differential TMV test structure. (b) Backside of the TMV test structure (c) Measured and simulated differential S-parameters from DC to 50 GHz showing good agreement.

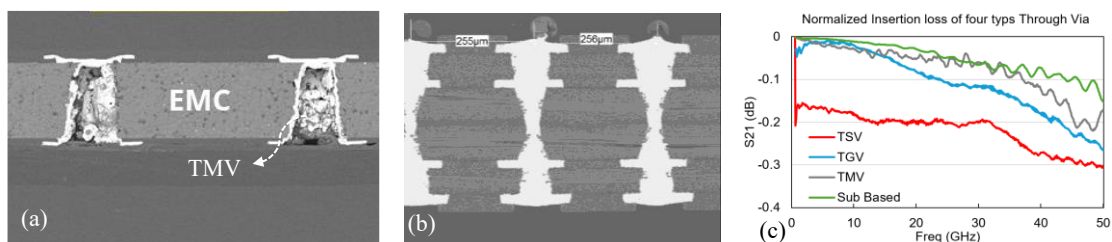


Figure 11. (a) Cross-sectional image of the fabricated TMV (b) Prefabricated substrate through via. (c) Normalized insertion loss of four types of through via from DC to 50 GHz (normalized to 100 μm via length).

Figure 12 (a) shows the fabricated interconnect from OE to switch through frontside RDL, TMV, C4 bumps, and substrate RDL. It was measured from DC to 60 GHz with an IL of 0.39 dB and 0.62 dB at 28 GHz and 56 GHz respectively as shown in Figure 12 (b). Figure 12 (c) shows the measured eye diagram (without pre-emphasis or equalization) for 112 Gbps PAM4 signal achieving an excellent TDECQ of 1.62 dB.

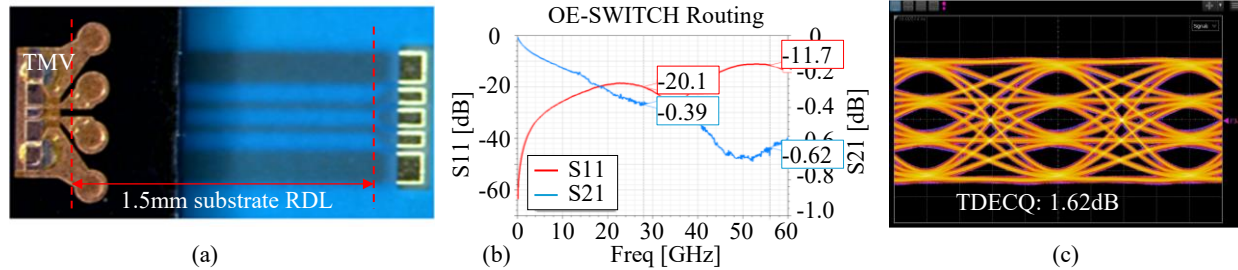


Figure 12. (a) OE to switch interconnects. (b) Measured differential insertion loss and return loss from DC to 60GHz. (c) Measured eye diagram for 112 Gbps PAM4 signals with a measured TDECQ of 1.62 dB.

4.3 Mechanical Performance and Reliability

The package has a predicted warpage of 19.5 μm . The package was modelled with dielectric layers (30 μm on frontside and 20 μm on backside) having a CTE of 65ppm/ $^{\circ}\text{C}$. Epoxy mold compound is cured at a temperature of 150 $^{\circ}\text{C}$ and therefore, its stress-free reference temperature. Dielectric polyimide layers are cured at 225 $^{\circ}\text{C}$ and solder reflow at 220 $^{\circ}\text{C}$. The embedded PIC tensile stress can be seen from the Figure 13 (a). After subjecting the assembly to temperature cycling tests ranging from -40 $^{\circ}\text{C}$ to 125 $^{\circ}\text{C}$, with a cycle duration of 1 hour, the predicted solder joint fatigue life of 1.6T OE is >4000 cycles.

4.4 Thermal performance

As shown in Figure 13 (c), a thermal simulation is carried out to depict the chip top surface temperature of OE with an airflow temperature of 25 $^{\circ}\text{C}$ and a speed of 5 m/s, employing a heatsink with a fin height of 10 mm. Temperature of 65 $^{\circ}\text{C}$ or less can be maintained in the PIC and therefore no adverse effect on its performance.

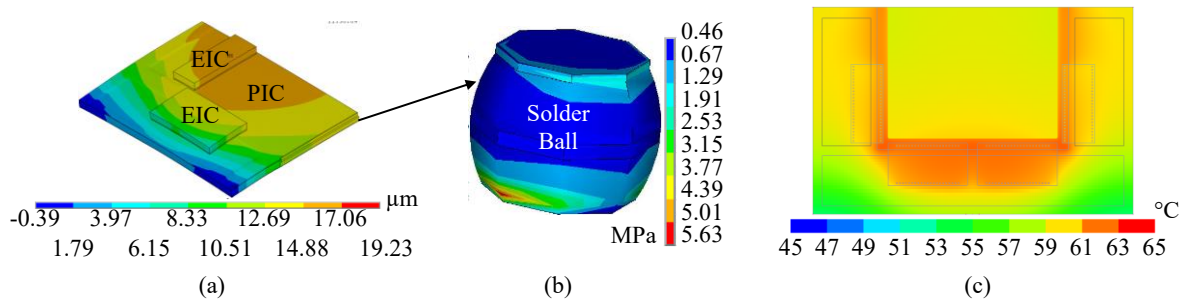


Figure 13. (a) A predicted warpage of 19.5 μm for 1.6T OE based on half symmetry mechanical model. (b) Strain energy density plot in first failure solder bump. (c) Thermal simulation of the OE package showing PIC temperature being maintained below 65 $^{\circ}\text{C}$ for 1.6T OE.

5. OPTICAL ENGINES FOR 6.4 TBPS AND BEYOND

Figure 14 (a) shows the 6.4 Tbps OE design with 32 I/O channels (224 Gbps $\times$ 32 for 6.4 Tbps or 448 Gbps $\times$ 32 for 12.8 Tbps). 8 EICs are flip-chip mounted on top of PIC. Prefabricated TGVs with fine pitch were utilized for the OE I/O paths to package substrate as shown in Figure 14 (b). Figure 14 (c) shows the simulated differential RL and IL from DC to 150 GHz. An IL of 0.05 dB and 0.09 dB is obtained at 56 GHz and 112 GHz respectively. Figure 15 shows the simulated eye diagrams of the TGV for 224 Gbps and 448 Gbps PAM4 signals with clear eye opening showing the OE interconnects' capability of carrying signals with 224 Gbps/lane and future 448 Gbps/lane. Figure 16 (a) shows a predicted warpage of 29 μm and Figure 16 (b) shows a PIC temperature of less than 65 $^{\circ}\text{C}$ is maintained for the 6.4T OE.

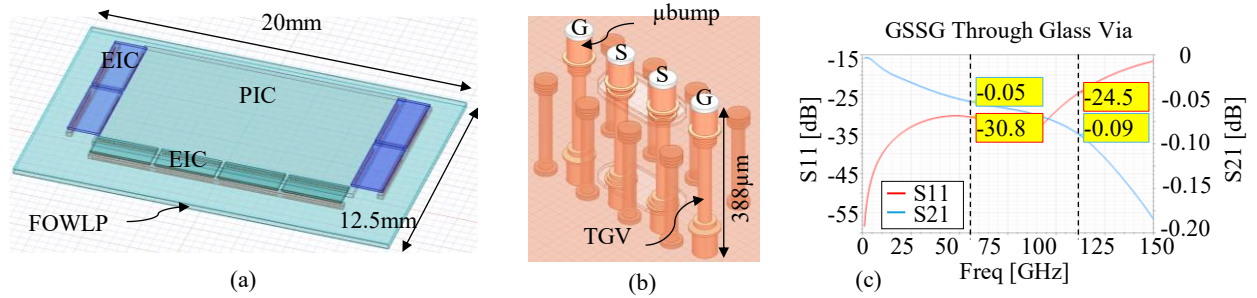


Figure 14. (a) 3D model of the OE package with 4 TIAs and 4 Drivers. (b) GSSG TGV interconnects from OE package I/O to package substrate. (c) Simulated differential S11 and S21 of the GSSG TGV structure.

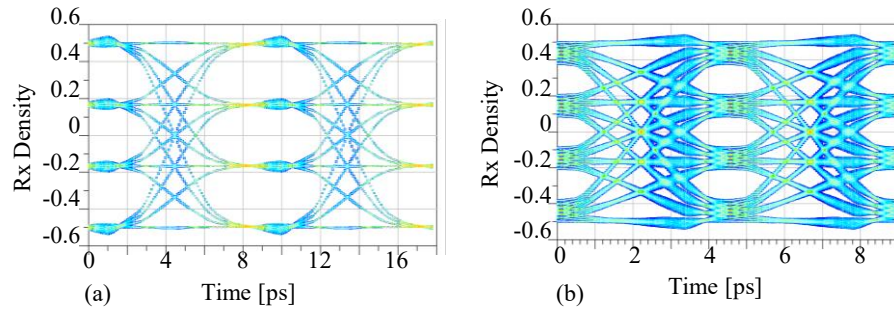


Figure 15. Simulated eye diagrams of the GSSG TGV structure for (a) 224 Gbps and (b) 448 Gbps PAM4 signals without any pre-emphasis or equalization.

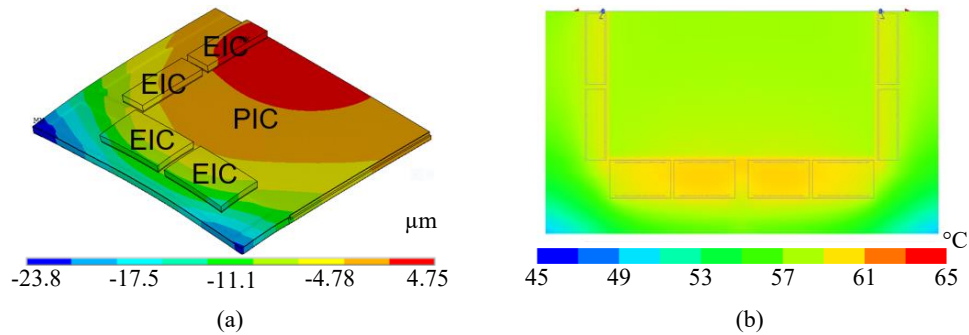


Figure 16. (a) A predicted warpage of 29 μ m for 6.4T and beyond OE. (b) Thermal simulation of the OE package showing PIC temperature being maintained below 65 $^{\circ}$ C for 6.4T and beyond OE.

6. CONCLUSION

We have presented a cost-effective, manufacturable, supply chain friendly heterogeneous integration platform using FOWLP for 1.6T/6.4T and beyond OE for hyperscale ML/AI and DC. In comparison to 3D-TSV and 2.5D interposer, FOWLP has both cost (50% lower) and performance benefits. 224 Gbps/ λ PAM4 signal routing capability of the interconnects was experimentally demonstrated with excellent optical eye TDECQ of 2.25 dB. Stress and thermomechanical simulations show that OE package warpage is less than 25 μ m and that the OE hot spot temperature is less than 65 $^{\circ}$ C. The presented FOWLP OE (9mmx13mm) demonstrates an aggregate 1.79 Tbps (224 Gbps $\times$ 8 channels) transmit and receive capability. We also show that the package can be scaled up to 20mmx20mm, and the interconnects are capable of carrying 448 Gbps/lane. With the larger package, the number of channels can be scaled up to 16 or 32 so that the OE can be scaled up to 3.2/6.4Tbps bandwidth or higher for future AI/ML clusters and hyperscale DCs.

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REFERENCES

- [1] S. Razdan, P. De Dobbelaere, J. Xue, A. Prasad and V. Patel, "Advanced 2.5D and 3D packaging technologies for next generation Silicon Photonics in high performance networking applications," *2022 IEEE 72nd Electronic Components and Technology Conference (ECTC)*, San Diego, CA, USA, 2022, pp. 428-435.
- [2] R. Mahajan *et al.*, "Co-Packaged Photonics For High Performance Computing: Status, Challenges And Opportunities," in *Journal of Lightwave Technology*, vol. 40, no. 2, pp. 379-392, 15 Jan.15, 2022.
- [3] R. Nagarajan *et al.*, "2.5D Heterogeneous Integration for Silicon Photonics Engines in Optical Transceivers," in *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 29, no. 3: Photon. Elec. Co-Inte. and Adv. Trans. Print., pp. 1-10, May-June 2023.
- [4] S. B. Nair Gourikutty *et al.*, "A Heterogeneously Integrated Wafer-level Processed Co-Packaged Optical Engine for Hyper-scale Data Centres," *2023 IEEE 73rd Electronic Components and Technology Conference (ECTC)*, Orlando, FL, USA, 2023, pp. 207-211.
- [5] X. Li *et al.*, "1.6 Tbps (224 Gbps/ λ) Silicon Photonic Engine Fabricated with Advanced Electronic-Photonic FOWLIP for Co-Packaged Optics and Linear Drive Applications," *2024 Optical Fiber Communications Conference and Exhibition (OFC)*, San Diego, CA, USA, 2024, pp. 1-3.
- [6] Jonghyun Cho and J. Kim, "Signal integrity design of TSV and interposer in 3D-IC," *2013 IEEE 4th Latin American Symposium on Circuits and Systems (LASCAS)*, Cusco, Peru, 2013, pp. 1-4.
- [7] W. Jiaqi, L. T. Guan, J. Tsung-Yang Liow and S. B. Nair Gourikutty, "Characterization of Differential TMV Vertical Interconnects to 50GHz with Double Side Measurement," *2023 IEEE 25th Electronics Packaging Technology Conference (EPTC)*, Singapore, 2023, pp. 664-668.